

When Brexit Means Brexit: Recovering Trade Policy Expectations from United Kingdom Import Stockpiling ^{*}

George Alessandria[†], Shafaat Yar Khan[‡] and Joseph Lansley[§]

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Abstract

We recover the evolution of expectations of future trade policy between the United Kingdom and Europe around the implementation of Brexit. Key to our analysis is a model of inventory management in the presence of future trade policy risk and trade flows between the United Kingdom and the rest of the world and Europe. We find that the first deadlines were viewed as more credible than the middle deadlines. We find that even in the month of Brexit agents put a non-trivial probability that Brexit would lead to no increase in United Kingdom trade barriers on imports from Europe.

JEL Classifications: F12, F13, F14

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[†]george.alessandria@rochester.edu, University of Rochester and NBER

[‡]shafaatyar.k@gmail.com, Syracuse University

[§]jlansley@UR.Rochester.edu, University of Rochester

1 Introduction

If trade policy is expected to change in the near future, firms have an incentive to shift the timing of their purchases to minimize the total cost of importing including tariffs, trade costs, and inventory holding costs. This incentive can give rise to large swings in trade without any current change in policy. These large swings in trade can be used to infer the expected change in future policy. In this paper, we use import dynamics into the UK from Europe and the RoW in the lead up to Brexit, with particular attention to the periods around the deadlines that came and went without and with Brexit, and a model of an importer importing subject to inventory frictions and uncertain future trade policy, to recover the time varying expectations of the likelihood of Brexit.

Brexit is a useful case to study the effect of an uncertain future change in trade policy on current trade flows for several reasons. First, the referendum set in motion a path that would lead to a change in trade barriers on United Kingdom trade with the EU but not with other trading partners. Second, in the period following the United Kingdom's referendum, the framework for Brexit had explicit deadlines - there were in total four commitments to resolve the exit process. This implies three revised commitments, each of which was undertaken a matter of days before additional trade costs were to have been imposed. Given the lack of legal precedent for EU departure, coupled with the disagreement in United Kingdom parliamentary, the episode featured limited institutional capacity for commitment. The Brexit deadlines involved a repeated trade policy threat, affecting a large and stable share of United Kingdom trade, but varying over time in perceived severity and credibility.

Our findings can be summarized in four related components: (i) there were significant responses in aggregate United Kingdom imports in anticipation of possible Brexit departure dates; (ii) the import data is well summarized by a $(\underline{s}, \bar{s})$ model of inventory accumulation with Markov trade policy; (iii) such patterns of inventory accumulation imply sizable perceived Brexit risk, which declined following the first deadline extension; and (iv) the *credibility* of deadline commitments, while initially declining, is ultimately highest at the final deadline.

We find evidence of significant anticipatory increases in imports ahead of proposed Brexit

deadlines. Our observation of a significant aggregate trade response to Brexit in the pre-Trade and Cooperation Agreement (TCA) period rigorously verifies patterns already visible in the raw data (ONS, 2021). Nonetheless, it stands in contrast with a large literature finding aggregate trade did not respond to Brexit prior to the formal departure date (Dhingra and Sampson, 2022; Freeman et al., 2025). This discrepancy results from our focus on monthly data. Aggregated to quarterly frequencies or lower, such effects are no longer visible. Our estimates highlight the important distinction between a general lack of responsiveness to uncertain future trade policy, and a rational absence of *low-frequency* intensive margin adjustment prior to the imposition of new trade costs.

To interpret these anticipatory dynamics, we deploy a standard $(\underline{s}, \bar{s})$ model of inventory accumulation (Alessandria et al., 2010) with multiple forms of trade policy and idiosyncratic shocks to fixed order costs. In this framework, importing firms use a storage technology allowing them to retain stock over time, subject to holding costs. In the face of a possible future trade cost increase, inventory accumulation drives import spikes like those we observe prior to Brexit deadlines. The magnitude of import growth closely mirrors perceived probability of trade costs materializing. Outside of considerable volatility in the course of the COVID-19 trade disruption, our dynamic model provides a close fit to the import data. An interpretation based on forward-looking stockpiling behavior provides a clean rationalization of empirical facts.

Utilizing our model’s implied relationship between trade cost probabilities and observable import magnitudes, we recover estimates of perceived Brexit risk among United Kingdom importers from the boom-bust cycles of imports surrounding the deadline periods. We find that each deadline introduced significant Brexit risk resulting in inventory accumulation. This risk declined considerably following the initial failed deadline of March 2019, suggesting the threat of a ‘disorderly’ Brexit was permanently reduced.

Finally, we deploy a novel identification approach to infer the credibility of Brexit deadlines. In addition to estimating the relative likelihood of ‘hard’ and ‘soft’ Brexit outcomes, we estimate the perceived likelihood of unresolved uncertainty, permitting a lack of credibility in the deadline commitments. Intuitively, we find that credibility declines by an order of magnitude following the failure to meet the first Brexit deadline. However, unlike the

perceived magnitude of Brexit risk, which remained lower through all subsequent deadlines, we find that credibility was recovered ahead of the final deadline commitment, implying deadline risk and deadline credibility were not perfectly correlated.

Brexit as a natural experiment comes with important limits to identification. Perceived Brexit costs, as opposed to Brexit *likelihood*, are not separately identified in our framework. We adopt an assumption of perfect foresight, in that firms perceive at each deadline a cost of Brexit consistent with the true reduction in trade following the TCA. Our estimates of first-moment Brexit probability are therefore best interpreted as a composite measure of Brexit risk, reflecting variation in both the likelihood and severity of trade costs.

1.1 Related Literature

A large literature documents a significant decline in United Kingdom imports from the EU following the TCA, albeit with varying magnitudes (Kren and Lawless, 2024; Freeman et al., 2025; Novy et al., 2024). These changes in United Kingdom trade were forecast by well-publicized ex-ante studies during the post-referendum period (Dhingra et al., 2017).

Despite consensus that Brexit implied a significant increase in trade costs, evidence for the pre-TCA period is equivocal. Crowley et al. (2019) find that the referendum itself had negative cross-sectional impacts on entry and exit decisions among United Kingdom firms. Graziano et al. (2021) and Fernandes and Winters (2021) similarly find cross-sectional impacts on the intensive margin. Nonetheless, Dhingra and Sampson (2022) and Freeman et al. (2025) demonstrate an absence of significant aggregate trade adjustments emerging from this cross-sectional variation over the four-year post-referendum period. Related studies reach similar conclusions in general equilibrium settings. Steinberg (2019) demonstrates the muted effect on trade of an increase in the second moment of policy uncertainty ahead of Brexit. Broadbent et al. (2024), meanwhile, explore the anticipatory post-referendum dynamics in a small-open-economy setting and find countervailing forward-looking dynamics working against a simple decline in United Kingdom-EU trade, distinct but complementary to the channel we consider.

Our results expand on this literature by providing evidence of high-frequency anticipatory behavior among firms. These findings are consistent with the absence of low-frequency

adjustment documented in both reduced-form and structural accounts, since the stockpiling incentives we identify are not observable once time periods are aggregated. Our model emphasizes the role of boom-bust dynamics in trade ahead of future trade costs, thereby reconciling the lack of low-frequency aggregate adjustment with forward-looking behavior.

An influential literature argues that Brexit-related uncertainty had large macroeconomic consequences outside of trade. Bloom et al. (2019) show that this uncertainty had large effects on United Kingdom aggregate investment, while Hassan et al. (2024) extend this methodology to address global uncertainty. We provide a consistent narrative regarding the time-variation of Brexit uncertainty, using a distinct identification approach. Furthermore, our evidence of inventory accumulation incentives can explain the co-existence of noticeable domestic macroeconomic effects with no notable trade adjustment: firms keenly anticipated Brexit costs, but for that reason they did not reduce trade prior to the imposition of costs.

Beyond the Brexit context, we contribute to a wider literature on Trade Policy Uncertainty (TPU) that begins with Handley and Limão, 2017 and Ruhl, 2011. Our results relate most directly to Alessandria et al. (2024) who study US product-level import patterns associated with the annual renewal of China’s MFN status. Similar to here, they find similar high-frequency dynamics of inventory accumulation and decumulation surrounding trade policy deadlines. We extend their framework to demonstrate the importance of this mechanism in another notable episode of repeated deadlines. A key difference in policy uncertainty here is that the policy deadlines were not always very spread apart and thus could interact with each other. Different from the China MFN renewal context is the Khan and Khederlarian (2021), in turn, find the converse property predicted by our model, with US importers running down inventories in anticipation of tariff *reductions* under NAFTA. Békés et al. (2017) find complementary evidence of the role of inventories in precautionary sales management, focusing on demand-side rather than supply-side volatility. Our paper also relates to recent work that uses trade flows and a dynamic exporting model to recover trade policy expectations (Alessandria et al., 2025b, Alessandria et al., 2025a, and Hoang and Mix, 2023). Our paper also relates to recent work that considers how inventories can be used in smoothing out shocks to trade costs (Alessandria et al., 2010 and Lafrogne-Joussier et al., 2023).

By providing a mechanism for the lack of a TPU-related decline in trade ahead of Brexit, we contrast with the view that TPU reduces trade in itself (Pierce and Schott, 2016; Handley and Limão, 2017). Our results suggest large time-varying Brexit uncertainty. The lack of an aggregate trade reduction therefore suggests direct negative effects of uncertainty on trade are dominated by dynamic substitution away from future trade costs.

1.2 Outline

The rest of the paper is organized as follows. In Section 2, we provide reduced-form analysis of United Kingdom import dynamics in the pre-TCA period. In Section 3, we introduce a structural model of inventory accumulation to estimate deadline-specific Brexit risk. In Section 4, we extend this model to allow firms’ expectations to doubt the credibility of Brexit deadlines, and estimate the perceived likelihood of uncertainty resolution. Section 5 provides some final thoughts.

2 Pre-Brexit Import Dynamics

We start by describing the dynamics of trade and trade policy in the period leading up to Brexit. Numerous economic analyses have suggested there was anticipatory stockpiling ahead of Brexit negotiation deadlines in import and export data (ONS, 2021). We deploy a formal econometric framework to measure the swings in trade flows around deadlines. We find evidence that United Kingdom imports from the EU displayed a significant increase ahead of each proposed deadline, after controlling for United Kingdom demand-side fixed effects and good-level seasonalities. We further find that both the magnitude and dynamics of anticipatory stockpiling vary by deadline.

2.1 Timeline of TCA Negotiations

We orient our empirical analysis around the negotiation timeline, which we now briefly summarize.

The United Kingdom voted to leave the European Union by referendum in June 2016, while the agreement on the resulting TCA was only reached four and a half years later, in December 2020. The period between these two events featured four policy commitments to final Brexit deadlines, and three attempts to credibly update prior commitments. In each

case, the validity or revision of the commitment was not revealed until several days before the stated horizon.

In March 2017, the United Kingdom triggered Article 50 of the Treaty on European Union, committing to leave the bloc within two years. In March 2019, eight days prior to the implied deadline, the United Kingdom and European Council agreed to rescind this commitment. The Council declared a state-contingent extension: countries would revert to MFN on April 12th without United Kingdom parliamentary approval for a Brexit deal.

On April 10th, the Council withdrew the second Brexit deadline, extending negotiations to October 31st 2019, despite the lack of a deal in United Kingdom parliament. The October deadline commitment remained in place until three days prior to reversion to MFN, on October 28th.

Policymakers proceeded to update their deadline commitment to January 1st 2021. Between the third and fourth deadlines, a United Kingdom general election delivered a landslide majority to the pro-Brexit government of Boris Johnson, while the United Kingdom entered the formal withdrawal process from the EU, without any indication of the trade agreement to be implemented.

On December 24th 2020, the final TCA deal was agreed, with volatile negotiations having continued up to this date. The associated changes in trade policy began implementation eight days later.

2.2 Aggregate Data

We start with a high level overview of trade dynamics with aggregate monthly United Kingdom import data from HMRC Overseas Trade Statistics. Throughout, we restrict our focus to the intensive margin of United Kingdom imports, including only goods which are traded at least once per year in our sample.

We emphasize that there appears to be noticeable differences in imports by source in periods around Brexit deadlines. Figure 1 presents aggregate United Kingdom imports from EU and the rest of the world, relative to October 2018. The four Brexit deadlines are indicated with dashed vertical lines. Prior to the first deadline in March of 2019, we see that United Kingdom imports from the EU and the RoW track extremely well. Only in February

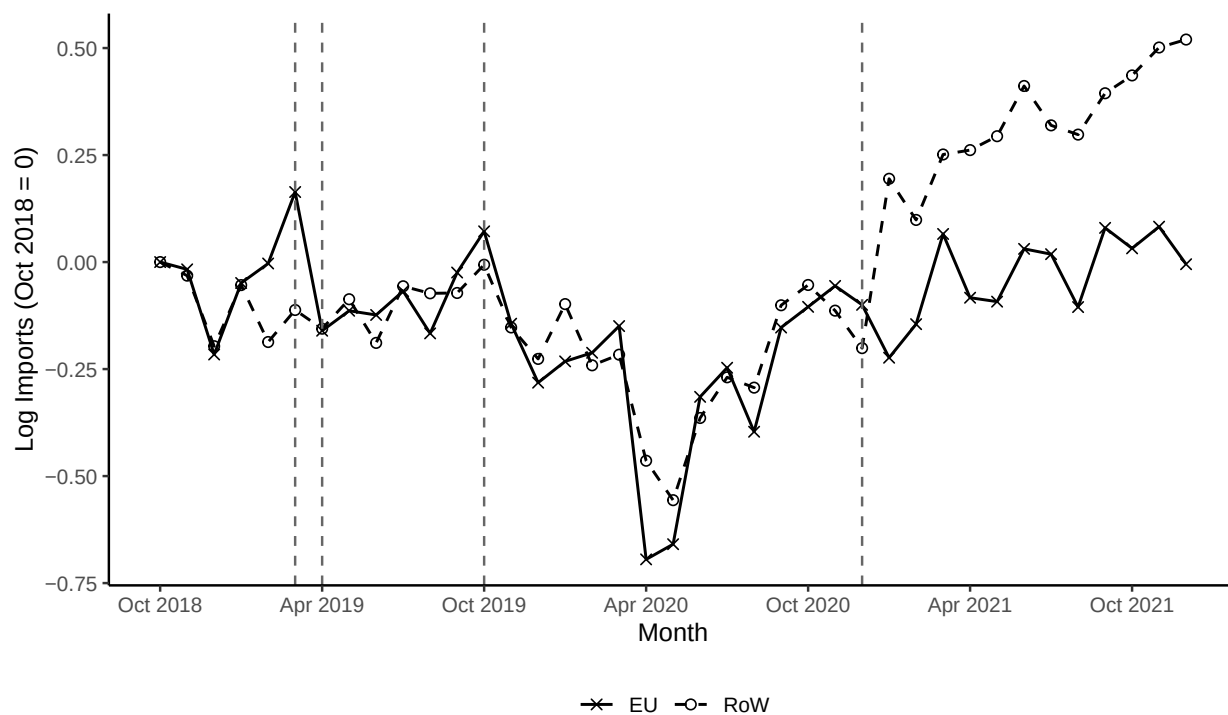


Figure 1: United Kingdom Imports from EU and RoW

Notes: Periods containing deadlines are marked with vertical dashed lines. Shaded regions are 95% confidence intervals. Block-shaded region indicates periods excluded from our structural estimation exercise. Source: HMRC Overseas Trade Statistics.

of 2019, do these series move apart. There is a particularly sharp increase in EU imports relative to RoW imports is visible ahead of the first deadline, with United Kingdom imports from the EU outstripping those from the rest of the world for two periods. For April 2019 EU and RoW are quite similar. The third and final deadlines each show a similar pattern to the first, but of considerably smaller magnitude.

Following the implementation of the TCA in January 2021, the two series diverge. Imports from the rest of the world display sustained growth, while imports from the EU remain largely stable.

2.3 Reduced-Form Specification

To improve our understanding of the suggestive patterns visible in the aggregate data, we shift to a reduced-form analysis of the underlying trade flows. This approach allows us to sort out whether there is just something unusual about the composition of trade around the Brexit deadlines or whether it is related to some perceived policy changes. Specifically, we aggregate trade flows to the HS6 level, defining each HS6 category as a good with generic index g . We aggregate exporting regions r to simply EU and RoW . Periods are indexed by year t and calendar month m . To abstract from extensive margin adjustments, we restrict our analysis to the subset of goods which are imported from both regions at least once per year in our sample.

To extract the differential growth each period of United Kingdom imports from the EU relative to the RoW , we estimate the following model:

$$\log V_{grtm} = \sum_{t'=2015}^{2020} \sum_{m'=1}^{12} \beta_{tm} \mathbf{1}\{t' = t \wedge m' = m \wedge r = EU\} + \gamma_{grm} + \gamma_{gtm} + \epsilon_{grtm}. \quad (1)$$

On the left-hand side, V_{grtm} denotes the value of United Kingdom imports indexed by good, region, year, and month. On the right, γ_{grm} controls for seasonal effects at the country-good level, while γ_{gtm} controls for United Kingdom demand shocks at the good level. The resulting coefficients, β_{tm} , extract the average EU-specific change in trade across goods in a month of our sample after controlling for seasonality.

In estimating Equation 1, we take 2015 as the reference year, as subsequent years are

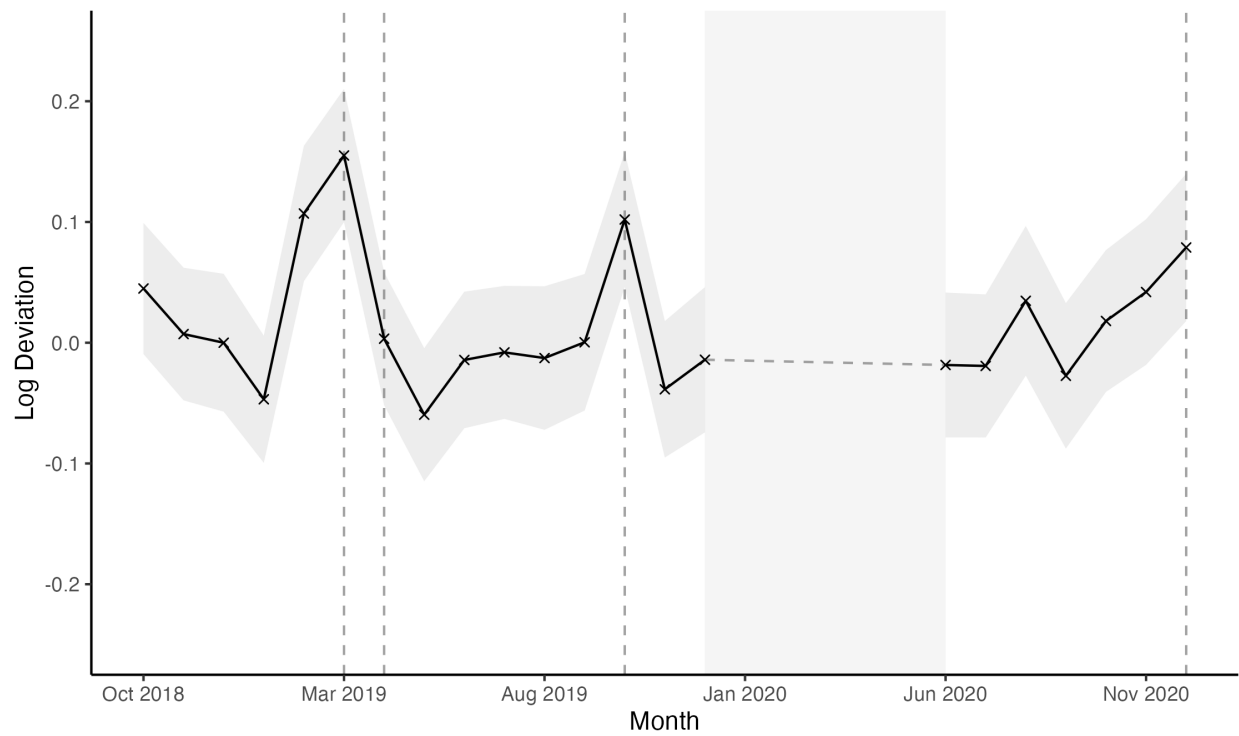


Figure 2: United Kingdom Excess Imports from EU

Notes: Periods containing deadlines are marked with vertical dashed lines. Shaded regions are 95% confidence intervals. Block-shaded region indicates periods excluded from our structural estimation exercise. Source: HMRC Overseas Trade Statistics.

liable to be impacted by the 2016 referendum. Figure 2 plots the import coefficients in our period of interest. Vertical lines indicate months including a proposed Brexit deadline, and shaded regions indicate 95% confidence intervals. Our subsequent analysis abstracts from bilateral volatility during the first half of 2020 on account of the COVID-19 pandemic, as marked in Figure 2 by a vertically shaded component.¹

There are three notable features of the resulting import dynamics. First, all except the April 2019 deadline exhibit a significant increase in imports above zero. This provides reduced-form evidence of anticipatory stockpiling behavior. Second, the pre-deadline import increases are heterogeneous in magnitude, with the first deadline’s peak increase in imports considerably exceeding that of the latter two. And third, the dynamics of adjustment differ across deadlines. The October 2019 deadline is characterized by only a one-period, sharply convex adjustment, with import behavior otherwise flat. Meanwhile, the first and final deadlines are spread over multiple periods, suggesting a more gradual increase.

These fluctuations imply United Kingdom importers responded in advance of future policy commitments regarding Brexit deadlines. This contrasts with a common view in the literature that observable Brexit implications are largely absent from the pre-TCA period (Dhingra and Sampson, 2022). The discrepancy arises due to the dimension of disaggregation: whereas previous work has emphasized sectoral or firm-level heterogeneity, the variation of interest here emerges in the distribution over time.

2.4 Sensitivity

We consider many alternative specifications and find our boom and bust around Brexit deadlines to be robust. Specifically, we introduce a measure of MFN tariffs and find essentially the same behavior. Moreover, we do not find any statistical significance of MFN rates interacted with deadlines, suggesting that MFN was not perceived to be the likely outcome of Brexit. We get quite similar results with a panel that starts sooner. Also, eliminating the seasonal controls does not influence our estimates around the first three deadlines. We find similar differential boom bust structure around deadlines in United Kingdom export data.

¹The finding that bilateral COVID-era United Kingdom-EU import disturbances are manifest after adjusting for the relevant fixed effects is also found by Freeman et al. (2025) using firm-level data.

3 Benchmark Structural Model

We now relate the import dynamics around Brexit deadlines documented to the dynamics of importer beliefs using a partial-equilibrium $(\underline{s}, \bar{s})$ model of inventory dynamics. Rationalizing the data requires a framework in which firms respond to the prospect of future trade policy changes by adjusting the timing of their imports. Faced with a proposed deadline with a positive probability of an increase in future trade costs have an incentive to front-load imports and hold larger inventories, thereby endogenously generating the large expansions and sharp contractions in imports observed in the data.

In this section, we outline the structure of a model we use to interpret the data. We do this in two steps. First, we develop the $(\underline{s}, \bar{s})$ inventory model in a stationary environment with constant trade policy. This model is closely related to the inventory model developed in Alessandria et al., 2010 extended to include an additional shock to fixed import costs, to capture challenges firms may face in restocking. Second, we introduce trade policy risk. A variation of the AKM $(\underline{s}, \bar{s})$ inventory model with a fixed policy risk has been studied by Alessandria et al., 2024. Here, we consider multiple uncertain reforms at irregular intervals. Then, by leveraging plausible structural parameters from the literature, we identify time-varying expectations of the likelihood of Brexit consistent with the evidence from the previous section.

3.1 Firm's Problem

There exists a continuum of importers, each selling a differentiated good. Time is discrete. At the beginning of period t , a firm i enters with an inventory stock $s_{it} \in \mathbb{R}_+$ of the good it imports and sells.

The firm faces two sources of shocks. First, there is a shock to demand and a shock to the fixed cost of importing. Conditional on this realization, the firm faces a demand curve

$$\tilde{q}_{it} = e^{z_{it}} p_{it}^{-\sigma}, \quad (2)$$

where σ denotes the demand elasticity. Actual sales are constrained by available inventories,

such that realized sales are

$$q_{it} = \min\{s_{it}, \tilde{q}_{it}\}. \quad (3)$$

Second, the firm draws an i.i.d. log-normally distributed demand shock $z_{it} \sim N(0, \zeta)$. With probability $1 - \xi$, the firm draws the finite fixed cost $\bar{f}$ and may re-stock in equilibrium. A re-stocking firm selects an import quantity $m_{it} \in \mathbb{R}_+$, incurring the fixed cost $\bar{f}$ and a variable cost of imports κ_t . Imports arrive in the subsequent period. The law of motion for inventories in this case is

$$s_{i,t+1} = (1 - \delta)(s_{it} + m_{it} - q_{it}), \quad (4)$$

where we impose the same depreciation rate on goods in transit.

Firms maximize the expected present discounted value of profits under risk neutrality, using a constant discount factor $\frac{1}{1+r}$. It is instructive to consider the case of a constant import cost, $\{\kappa_t\}_{t=0}^\infty = \bar{\kappa}$. Under this restriction, optimal pricing and import decisions admit a stationary recursive characterization.

A firm's state is defined by its inventory s , demand shock z , and fixed cost draw f . Let $\iota \in \{0, 1\}$ denote the discrete choice decision to re-stock, such that $\iota = 0$ imposes $m = 0$. The firms dynamic programming problem is

$$\begin{aligned} V(s, z, f) = \max_{p, m, \iota} & \left\{ pq(p, s, z) - \iota(f + \bar{\kappa}m) + \frac{1}{1+r} \mathbb{E}[V(s', z', f')] \right\} \\ \text{s.t. } & s' = (1 - \delta)[s - q(p, s, z) + \iota m], \end{aligned} \quad (5)$$

In this Markovian setting, conditional on choosing to re-stock ($\iota = 1$), the firm optimally adjusts inventories to a constant target level $\bar{s}$. This target is implicitly defined by

$$\bar{\kappa} = \left(\frac{1 - \delta}{1 + r} \right) \mathbb{E} \left[\frac{\partial V}{\partial s}(\bar{s}, z', f') \right], \quad (6)$$

where independence of future z and f draws from current values implies that $\bar{s}$ is independent of the current state. Conditional on drawing a finite fixed cost, the discrete decision to import is more likely when demand is high or inventory are low.

3.2 Benchmark Belief Structure: Credible Deadlines

The pre-Brexit period was characterized by time-varying risks of future trade costs. To incorporate this, we generalize the static constant- κ environment to include shocks to κ at specified intervals that correspond with Brexit deadlines. Let $\mathcal{T}$ denote the set of months in the event-study window depicted in Figure 1, from November 2018 to December 2020. Define $\mathcal{D} \subset \mathcal{T}$ as the set of months containing deadlines, with four elements $\mathcal{D} = \{d_1, d_2, d_3, d_4\}$.

The mass of importers begin in the initial period with import cost κ_0 and a steady-state distribution of inventories λ_{SS} . At a deadline $d_i \in \mathcal{D}$, firms believe with deadline-specific probability η_i that they will face a permanent trade cost $\tilde{\kappa}$ beginning from the next period. Note that we assume that $\tilde{\kappa}$ is deadline invariant. All heterogeneity in imports across deadlines is driven by variation in the perceived *likelihood* of a hard Brexit, not variation in its perceived severity. As discussed in the Introduction, this identifying assumption leads to an interpretation of η_i as deadline i 's general Brexit probability.

In this benchmark specification, firms believe at each deadline that with probability $1 - \eta_i$, the import cost is permanently fixed at κ_0 . If a 'hard' Brexit is not realized, a 'soft' Brexit occurs with certainty. This assumption imposes credibility of the Brexit deadlines ex-ante, as firms believe uncertainty is guaranteed will be resolved at each deadline. In Section 4, we extend the model to allow agents to allow the deadline to be delayed.

In this environment, the importer dynamics discussed in Section 3.1 are shaped by a sequence of continuation values $\{V_t\}_{t \in \mathcal{T}}$, $V_t : S \times \mathbb{R} \times \{\bar{f}, \infty\} \rightarrow \mathbb{R}$. These in turn depend on hard- and soft-Brexit continuation values, $\{V_0, V_{\tilde{\kappa}}\}$. Here, V_0 is the static value function characterized by Equation 5 with $\bar{\kappa} = \kappa_0$, while $V_{\tilde{\kappa}}$ is the analogous value function with $\bar{\kappa} = \tilde{\kappa}$. Note, because there are permanent changes in trade costs these value functions do not need to be indexed by time.

We now write out the dynamic programming problem of an importer in time t prior to the realization of Brexit as

$$\begin{aligned} V_t(s, z, f) = \max_{p, m, \iota} & \left\{ pq(p, s, z) - \iota(f + \kappa_0 m) + \frac{1}{1+r} \mathbb{E}[V_{t+1}(s', z', f')] \right\} \\ \text{s.t.} \quad & s' = (1 - \delta)[s - q(p, s, z) + \iota m], \end{aligned} \tag{7}$$

noting that with constant pre-Brexit variable import cost κ_0 , a period's value function is fully determined by the subsequent period's continuation value.

Post-deadline continuation values are defined with respect to the hard- and soft-Brexit outcomes,

$$\mathbb{E}[V_{d_i}] = \eta_i V_{\bar{\kappa}} + (1 - \eta_i) V_0, \quad (8)$$

since the expected post-deadline continuation value is simply the expectation defined over hard- and soft-Brexit outcomes. Inter-deadline value functions are inferred by backward induction from the upcoming deadline commitment.

The resulting sequence of values can be mapped to the aggregate import quantities of our data by solving forwards for the inventory distributions. The sequence of value functions implies a sequence of re-stock policies $\{s'_t\}_{t \in \mathcal{T}}$, which in turn induce import policies $\{m_t\}_{t \in \mathcal{T}}$. Given a distribution over inventories at t denoted λ_t , we have

$$\lambda_{t+1}(s) = \int_S \int_{\mathbb{R}} [(1 - \xi) \mathbf{1}\{s'(u, z, \bar{f}) = s\} + \xi \mathbf{1}\{s'(u, z, \infty) = s\}] d\Phi(z) d\lambda_t(u) \quad (9)$$

for all $s \in S$, where Φ is the demand shock distribution. Beginning from the cost- $\bar{\kappa}$ steady-state distribution λ_{SS} , this recursion yields a sequence of distributions of inventories, $\{\lambda_t\}_{t \in \mathcal{T}}$. Imports at each date follow from the orders of firms with finite fixed order costs

$$M_t = (1 - \xi) \int_S \int_{\mathbb{R}} m(s, z, \bar{f}) d\Phi(z) d\lambda_t(s). \quad (10)$$

Conditional on structural parameters, this system defines a mapping from deadline belief parameters $\{\eta_i\}_{i=1}^4$ to aggregate import sequences $\{M_t\}_{t \in \mathcal{T}}$ comparable to the data.

3.3 Estimation

The event-study coefficients of Section 2.3 provide a sequence of moment restrictions $\{\beta_t\}_{t \in \mathcal{T}}$ identifying the importer expectations $\{\eta_i\}_{i=1}^4$.² We further deploy this sequence to discipline the Calvo importing parameter ξ .

To proceed with estimating these parameters of interest, we require externally set values

²Given the extreme deviations in imports in April-May 2020, we omit these moments in our estimation procedure.

for structural parameters $\{\sigma, \delta, r, \zeta, \bar{f}, \kappa_0, \tilde{\kappa}\}$. We choose r for an annual discount rate of 3%, and normalize κ_0 to deliver unit demand in steady state. Following Alessandria et al. (2010), we set $\delta = 5\%$ and $\bar{f} = 0.1$; following Khan and Khederlarian (2021), we set $\sigma = 2.7$ to be consistent with short-run trade elasticities from tariff changes.

In our core specification we choose $\tilde{\kappa}$ to match the observed decline in United Kingdom imports from the EU in the year following the TCA. In particular, given the externally set parameters outlined above, we choose cost $\tilde{\kappa}$ such that the corresponding steady state of the model features the same import reduction as the mean fitted 2021 elasticity.

Table 1: Externally Set Parameters

Parameter	Description	Source	Value
σ	Demand elasticity	Khan & Khederlarian (2021)	2.7
δ	Inventory depreciation rate	Alessandria et al. (2010)	5%
$\bar{f}$	Fixed cost of importing	Alessandria et al. (2010)	0.1
r	Interest rate	Std. time preference	3%
ζ	St. deviation of demand	Ad hoc	1
κ_0	Pre-TCA import cost	Normalized	$\frac{\sigma}{\sigma-1}$
$\tilde{\kappa}$	Post-TCA import cost	Post-TCA import elasticity	$1.09 \times \kappa_0$

Given these externally set structural parameters, a vector $\theta := \{\{\eta_i\}_{i=1}^4, \xi\}$ gives rise to an aggregate imports sequence $\{M_t(\theta)\}_{t \in \mathcal{T}}$ and steady state $M_{SS}(\theta)$ by Equations 8-12 such that we can define model-implied elasticities

$$\tilde{\beta}_t(\theta) = \log \frac{M_t(\theta)}{M_{SS}(\theta)} \quad (11)$$

for all $t \in \mathcal{T}$. We then obtain our estimate $\hat{\theta}$ by Simulated Method of Moments using the objective

$$\hat{\theta} = \arg \min_{\theta \in [0,1]^5} \left\{ (\beta - \tilde{\beta})' W (\beta - \tilde{\beta}) \right\} \quad (12)$$

where the weighting terms are simply the inverse of standard errors of event-study elasticities, $W = \text{diag}(\frac{1}{\text{s.e.}(\beta)})$.

3.4 Results

The first column of Table 2 summarizes the deadline-specific perceived Brexit risk we recover from this exercise. We find that the perceived risk associated with the initial deadline was an order of magnitude higher than subsequent cases. This aligns with an interpretation whereby the initial deadline delay revealed policymakers’ underlying aversion to a ‘disorderly’ hard Brexit, permanently reducing the probability of large trade costs.

Deadline	Credible Deadline Model	Extended Belief Structure	
	η	γ	η
d_1	15%	7%	14%
d_2	7%	0%*	7%
d_3	8%	93%	6%
d_4	8%	0%	8%

Table 2: Estimated Belief Parameters by Model Specification

Notes: * indicates normalized value. First column refers to estimation detailed in Section 3.3. Second and third columns refer to estimation detailed in Section 4. RMSE of benchmark model is 0.00127.

The values associated with subsequent deadlines are stable thereafter, suggesting limited direct adjustment to Brexit risk after the initial revision. The similar η value attached to the second deadline and its successors underscores the importance of a feasible structural model: while the April 2019 import deviation is near zero in the data, without positive Brexit risk at this date, stock accumulated in the March import surge would imply a large negative elasticity. By keeping track of a sequence of inventory distributions, λ in our notation, we can measure the persistent risk driving the anomalous absence of a post-deadline crater in our elasticities.

While our Brexit risk terms are non-trivially positive, their magnitudes are relatively low. Recall that $\tilde{\kappa}$, the import costs associated with a hard Brexit in our specification, are calibrated to match the import reductions of the realized TCA. Relative to contemporaneous discussions of the possible distribution of Brexit risk, this is a conservative description of the worst-case scenario.

Figure 3 plots the simulated elasticities associated with our estimated parameters against the targeted moments from the data. Outside of the untargeted COVID-19 period, the model provides a compelling fit of the core dynamics given the estimated parameters are four times

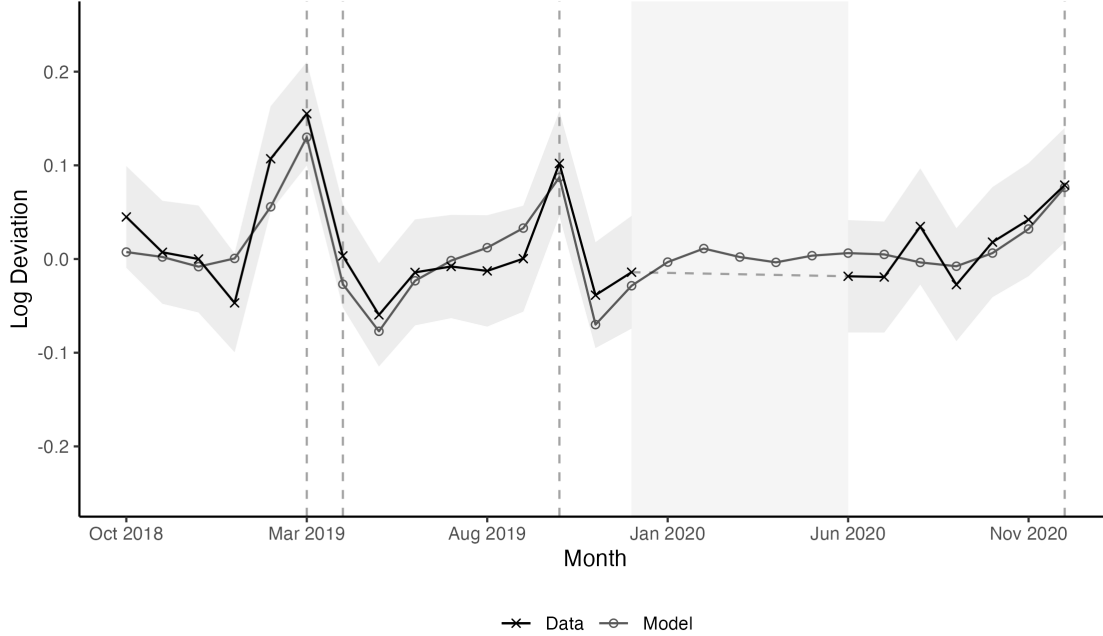


Figure 3: Benchmark Model Fit

Notes: Model-generated imports follow the estimation procedure in Section 3.3. Periods containing deadlines are marked with vertical dashed lines. Shaded regions are 95% confidence intervals. Block-shaded region indicates periods excluded from estimation.

over-identified. This further reinforces the view that forward-looking stockpiling behavior provides a parsimonious explanation for the fluctuations in aggregate United Kingdom trade in this period.

4 Estimating Deadline Credibility

Our benchmark belief specification imposed the credibility of the Brexit deadlines by default. Importers are sure that uncertainty will be permanently resolved at each threshold. Given the repeated adjustment of deadline commitments, as documented in Section 2.1, we relax this assumption. In this section, we extend the benchmark model to recover time-varying importer beliefs over the probability of uncertainty resolution.

4.1 Extended Belief Structure

The distribution of Brexit risk has so far been characterized by two value functions, V_0 and $V_{\tilde{\kappa}}$. Both of these value functions were stationary, in that they refer to environments with static fundamentals.

To extend this binary support with the minimum of additional complexity, we introduce a state of persistent risk codified by a value function V_ϵ . The continuation value under persistent risk is defined by the following recursion:

$$V_\epsilon(s, z, f) = \max_{p, m, \iota} \left\{ pq(p, s, z) - \iota(f + \kappa_0 m) + \frac{1}{1+r} \mathbb{E}[(1-\epsilon)V_\epsilon(s', z', f') + \epsilon V_{\tilde{\kappa}}(s', z', f')] \right\}$$

$$\text{s.t. } s' = (1-\delta)[s - q(p, s, z) + \iota m]. \quad (13)$$

In this case, agents do not face immediate trade costs post-deadline. But uncertainty remains unresolved, as with probability ϵ , Brexit costs are realized in the subsequent period.

Such an outcome reflects the experience of United Kingdom importers in the transition from the March 2019 to the April 2019 deadlines. Brexit did not occur, but a high level of imminent risk remained. It is therefore natural to suppose that the high-frequency adjustment of policy commitments amid political uncertainty may naturally have caused anticipation of multiple periods of disruption.

We maintain the structure of Section 3.2, with an additional parameter introducing the possibility of persistent risk. For $d_i \in \mathcal{D}$, let γ_i denote the probability of continued uncertainty. With probability $1 - \gamma_i$, the deadline proceeds as before: a hard Brexit occurs with probability η_i , and a soft Brexit with probability $1 - \eta_i$. At each deadline beliefs over the continuation value are distributed over three functions, $\{V_{\tilde{\kappa}}, V_0, V_\epsilon\}$, with respective probability weights $\{(1 - \gamma_i)\eta_i, (1 - \gamma_i)(1 - \eta_i), \gamma_i\}$.

4.2 Identifying γ

At a given deadline $d_i \in \mathcal{D}$, separate identification of η_i and γ_i is non-trivial. Both reflect the risk of future trade costs, increasing the marginal value of inventories and incentivizing precautionary stockpiling. The parameters are therefore substitutable in matching the peak and trough associated with a given deadline. Under the structure of a standard $(\underline{s}, \bar{s})$ inventory model, this largely exhausts the restrictions provided by the data.

As discussed in Section 3.4, however, the Brexit deadlines display multi-period anticipatory behavior, motivating our inclusion of shocks to the import costs, ξ . The possibility of being unable to import in a given period induces sensitivity to the distribution of trade

cost risk through time, breaking the observational near-equivalence between $\{\eta_i, \gamma_i\}$ combinations. Parameters $\{\eta_i\}_{i=1}^4$ pose immediate risk, leaving the firm highly exposed to import stickiness in the period of the deadline. This drives multiple periods of precautionary importing and induces concavity in pre-deadline imports. Risk of persistent uncertainty $\{\gamma_i\}_{i=1}^4$ may impose the same present-value effect on a given flow of imports at a particular date, but due to the dispersed nature of the risk, the effect will be very minimally affected by Calvo import stickiness. Given the low probability of trade costs occurring at any single date, multi-period anticipation of a low-credibility deadline is muted, yielding a sharply convex increase in the pre-deadline imports.

By expanding the model’s sensitivity beyond the deadline period, we are able to leverage information codified in our over-identifying restrictions which is otherwise lost. Figure 4 illustrates this method in a toy example. We fix two γ values, one with $\gamma = 0.0$ (full credibility) and one with $\gamma = 0.9$ (low credibility). Both scenarios have the same stickiness parameter ξ , and both have an η value adjusted to deliver the same deadline peak. The import level is higher for the credible deadline for two periods ahead of the deadline itself, with the penultimate import elasticity almost double that of the low-credibility deadline. As such, data on multi-period anticipatory stockpiling can be utilized in our model to identify deadline credibility.

4.3 Results

Having outlined our identification strategy, we now return to the empirical Brexit deadlines of the reduced-form analysis. As our approach relies directly on having multiple periods of anticipation, we cannot estimate a unique value for γ_2 (corresponding to the April 2019 deadline), and consequently we normalize it to zero.

As summarized in the third column of Table 2, the recovered γ estimates diverge considerably between deadlines. Intuitively, there is a considerable reduction in confidence following the failure to adhere to the first deadline. Whereas the two-year interval specified by EU legislation was deemed credible grounds for responding to the initial deadline, the precedent set by a first deviation in March 2019 resulted in very low confidence in the delayed deadline.

Between the third and final deadline, however, there is a reversal, and the final Brexit

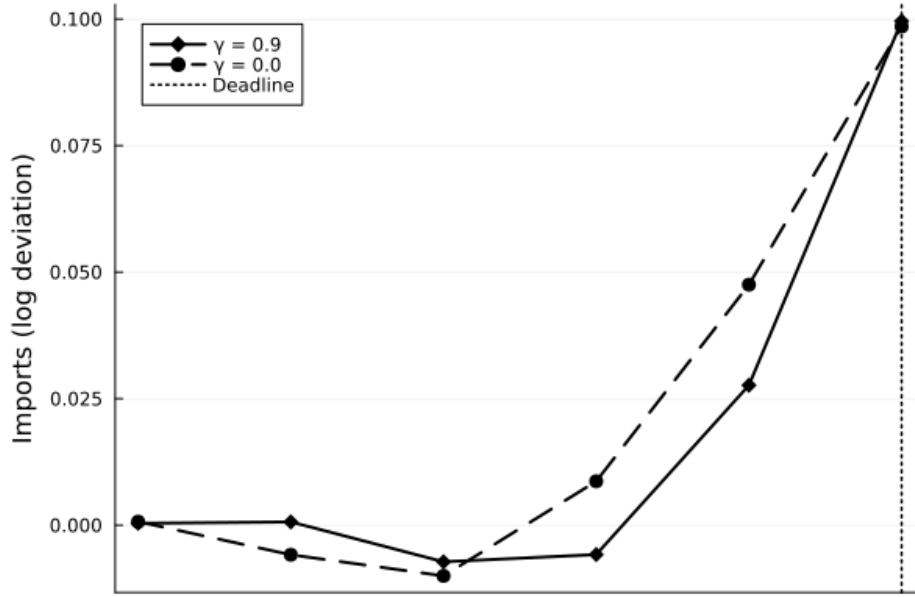


Figure 4: Numerical Example of Credible and non-Credible Deadline Anticipation

deadline was correctly recognized as more credible than the initial departure date. This may be interpreted as a responsiveness among importers to changes in United Kingdom political circumstances. The return to credibility was preceded by a landslide election victory for a campaign with a highly-publicized commitment to delivering Brexit. Our results cohere with a narrative that such a change in political circumstance may have reversed the near-total loss of credibility experienced by the negotiations process in 2019.

As discussed in Section 4.2, these conclusions are driven by the multi-period anticipation evident in our elasticities. The low degree of credibility at the October 2019 deadline is apparent from the sharp convexity of the import elasticities, with no visible import growth prior to the deadline period. In contrast to this, both the first and final deadline exhibit anticipatory import growth in the penultimate period. The increased credibility attributed to the final deadline is motivated by the two preceding periods, with a modest uptick in imports in October 2020 indicating the greatest degree of forward-looking inventory accumulation of the three directly anticipated deadlines.

A striking feature of our results is the lack of correlation between Brexit risk and Brexit credibility. As documented in the first column of Table 2, the March 2019 deadline continues to provoke considerably greater risk than the December 2020 case in the extended beliefs

specification. The return to credibility under the final deadline did not re-institute the severity of the threat of a deeply damaging outcome, exemplifying the distinction between expected trade costs and uncertainty proper which our model identifies.

5 Final Thoughts

We use the dynamics of United Kingdom imports and an inventory model with time varying trade policy risk to recover the dynamics of expected trade policy around the different Brexit target deadlines. Our analysis demonstrates how to recover repeated near-term trade policy risk using an inventory model. Similar analysis could be done on US trade flows in recent times as those have been characterized by substantial anticipatory imports and exports (Alessandria et al., [2025c](#)).

References

- Alessandria, George, Joseph P. Kaboski, and Virgiliu Midrigan (2010). “Inventories, Lumpy Trade, and Large Devaluations”. *American Economic Review* 100.5, pp. 2304–2339.
- Alessandria, George, Shafaat Yar Khan, Armen Khederlarian, Kim J. Ruhl, and Joseph B. Steinberg (2025a). “Trade War and Peace: U.S.-China Trade and Tariff Risk from 20152050”. *Journal of International Economics* 155, p. 104066. ISSN: 0022-1996. DOI: <https://doi.org/10.1016/j.jinteco.2025.104066>. URL: <https://www.sciencedirect.com/science/article/pii/S0022199625000224>.
- (2025b). “Trade-Policy Dynamics: Evidence from 60 Years of U.S.-China Trade”. *Journal of Political Economy*. DOI: [10.1086/733420](https://doi.org/10.1086/733420). eprint: <https://doi.org/10.1086/733420>. URL: <https://doi.org/10.1086/733420>.
- Alessandria, George A, Jiaxiaomei Ding, Shafaat Y Khan, and Carter B Mix (May 2025c). *The Tariff Tax Cut: Tariffs as Revenue*. Working Paper 33784. National Bureau of Economic Research. DOI: [10.3386/w33784](https://doi.org/10.3386/w33784). URL: <http://www.nber.org/papers/w33784>.
- Alessandria, George A., Shafaat Y. Khan, and Armen Khederlarian (2024). “Taking Stock of Trade Policy Uncertainty: Evidence from Chinas Pre-WTO Accession”. *Journal of International Economics* 150.
- Békés, Gábor, Lionel Fontagné, Balázs Muraközy, and Vincent Vicard (2017). “Shipment frequency of exporters and demand uncertainty”. *Review of World Economics* 153.4, pp. 779–807.
- Bloom, Nick, Philip Bunn, Stephen Chen, and Paul Mizen (2019). “Brexit and uncertainty: a new measure of uncertainty and its impact on UK investment”. *Fiscal Studies* 40.1, pp. 5–33.
- Broadbent, Ben, Federico Di Pace, Thomas Drechsel, Richard Harrison, and Silvana Tenreyro (2024). “The Brexit Vote, Productivity Growth and Macroeconomic Adjustments in the U.K.” *The Review of Economic Studies* 91.4, pp. 2104–2150.
- Crowley, Meredith A., Oliver Exton, and Lu Han (2019). *Renegotiation of Trade Agreements and Firm Exporting Decisions: Evidence from the Impact of Brexit on UK Exports*. Discussion Paper. Centre for Economic Policy Research (CEPR).

- Dhingra, Swati, Hiu Huang, Gianmarco I.P. Ottaviano, João Paulo Pessoa, Thomas Sampson, and John Van Reenen (2017). “The costs and benefits of leaving the EU: trade effects”. *Economic Policy* 32.92, pp. 651–705.
- Dhingra, Swati and Thomas Sampson (2022). “Expecting Brexit”. *Annual Review of Economics* 14.1, pp. 495–519.
- Fernandes, Ana P. and L. Alan Winters (2021). “Exporters and shocks: The impact of the Brexit vote shock on bilateral exports to the U.K.” *Journal of International Economics* 131.
- Freeman, Rebecca, Marco Garofalo, Kalina Manova, Rebecca Mari, Thomas Prayer, and Thomas Sampson (2025). “Deep Integration and Trade: UK Firms in the Wake of Brexit”. *CESifo Working Paper*.
- Graziano, Alejandro G., Kyle Handley, and Nuno Limão (2021). “Brexit Uncertainty and Trade Disintegration”. *The Economic Journal* 131.635, pp. 1150–1185.
- Handley, Kyle and Nuno Limão (2017). “Policy uncertainty, trade, and welfare: Theory and evidence for China and the U.S.” *American Economic Review* 107.9, pp. 2731–2783.
- Hassan, Tarek A., Stephan Hollander, Laurence van Lent, and Ahmed Tahoun (2024). “The Global Impact of Brexit Uncertainty”. *Journal of Finance* 79.1, pp. 413–458.
- Hoang, Trang and Carter B. Mix (2023). “Trade War, Adjustment Dynamics, and Expectations”. Mimeo.
- Khan, Shafaat Yar and Armen Khederlarian (2021). “How does trade respond to anticipated tariff changes? Evidence from NAFTA”. *Journal of International Economics* (133).
- Kren, Janez and Martina Lawless (2024). “How has Brexit changed EUUK trade flows?” *European Economic Review* 161.
- Lafragne-Joussier, Raphael, Julien Martin, and Isabelle Mejean (Mar. 2023). “Supply Shocks in Supply Chains: Evidence from the Early Lockdown in China”. *IMF Economic Review* 71.1, pp. 170–215. DOI: [10.1057/s41308-022-00166-8](https://doi.org/10.1057/s41308-022-00166-8). URL: https://ideas.repec.org/a/pal/imfecr/v71y2023i1d10.1057_s41308-022-00166-8.html.
- Novy, Dennis, Thomas Sampson, and Catherine Thomas (2024). *Brexit and UK trade*. Discussion Paper. Centre for Economic Policy Research (CEPR).
- ONS (2021). *Did UK firms stockpile items ahead of the Brexit deadline?*

- Pierce, Justin R. and Peter K. Schott (2016). “The Surprisingly Swift Decline of U.S. Manufacturing Employment”. *American Economic Review* 106.7, pp. 1632–1662.
- Ruhl, Kim (2011). “Trade dynamics under policy uncertainty”. *American Journal of Agricultural Economics: Papers and Proceedings* 93 (2), pp. 450–456.
- Steinberg, Joseph B. (2019). “Brexit and the macroeconomic impact of trade policy uncertainty”. *Journal of International Economics* 117, pp. 175–195.